

Real-Time Traffic Sign Recognition with Open-Set Rejection and Temporal Consensus

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Abstract—Traffic sign recognition (TSR) is a critical perception component for advanced driver assistance and intelligent transportation systems, where accuracy must be achieved under stringent real-time and deployment constraints. In unconstrained roadside environments, traffic signs often appear at small scales, undergo partial occlusion, and coexist with visually similar non-traffic objects, leading to frequent false detections and unstable predictions across video frames. Moreover, conventional TSR models are typically trained under closed-set assumptions, limiting their reliability when encountering unseen or region-specific signage.

This paper presents a real-time TSR framework designed for open-world video scenarios by jointly addressing false-positive suppression and temporal instability. The proposed system integrates a lightweight single-stage detector optimized for small-object localization with a hierarchical sign categorization strategy. To explicitly handle unknown or visually ambiguous signage, an open-set rejection mechanism is introduced, supported by low-cost geometric and color-consistency constraints derived from standard traffic sign design principles. In addition, a temporal consensus module aggregates predictions across tracked instances, improving recognition stability under motion blur and intermittent occlusions without increasing inference latency.

The complete pipeline is optimized for edge-level deployment and achieves sub-50 ms end-to-end inference on embedded GPUs. Experimental evaluation on mixed public benchmarks and real-world roadside video demonstrates improved precision, robust rejection of non-sign objects, and stable temporal predictions compared to conventional frame-wise TSR baselines. The proposed framework establishes a practical and extensible baseline for real-time open-world traffic sign recognition.

Index Terms—Traffic sign recognition, real-time detection, open-set recognition, temporal consistency, edge deployment

I. INTRODUCTION

Traffic Sign Recognition (TSR) is a fundamental perception component in Advanced Driver Assistance Systems (ADAS) and intelligent transportation systems, where accurate interpretation of roadside signage directly influences driving safety, regulatory compliance, and autonomous decision-making [1]. Unlike controlled object recognition tasks, TSR must operate reliably in unconstrained outdoor environments characterized by significant variation in scale, illumination, weather conditions, and background clutter. Traffic signs often occupy only a small fraction of the visual field and may be partially occluded or degraded due to motion blur, making robust real-time recognition particularly challenging [2], [3].

Recent advances in deep learning have substantially improved TSR performance through convolutional neural net-

works and end-to-end object detection frameworks. One-stage detectors, particularly YOLO-based architectures, enable joint localization and classification while satisfying real-time constraints [4], [5]. These approaches have demonstrated high accuracy on widely used benchmarks such as GTSRB and GTSDb [6], [7]. However, most existing TSR systems are developed under a closed-set assumption, where all test instances are expected to belong to a predefined and fixed set of known traffic sign categories.

In real-world driving environments, the closed-set assumption is frequently violated. Vehicles routinely encounter visually similar non-traffic objects such as advertisements, storefront signs, billboards, and temporary or region-specific signage that is absent from training data. Closed-set classifiers tend to assign such unknown inputs to the most visually similar known class with high confidence, resulting in false positives that may propagate incorrect information to downstream driving modules [8], [9]. From a safety perspective, such overconfident misclassification is often more problematic than missed detections, as it may trigger inappropriate driver alerts or control actions.

Another persistent challenge in video-based TSR systems is temporal instability. Frame-wise recognition pipelines process each frame independently and often fail to exploit the strong temporal continuity inherent in driving scenarios. As a result, transient effects such as motion blur, camera vibration, sudden illumination changes, or partial occlusion can lead to inconsistent predictions for the same physical sign across consecutive frames [10], [11]. These fluctuations manifest as prediction flicker or delayed recognition, which are undesirable in safety-critical applications and may reduce user trust in automated driving assistance systems.

From a deployment standpoint, TSR systems must also comply with strict real-time and resource constraints imposed by embedded and edge computing platforms. Automotive-grade hardware limits memory capacity, power consumption, and allowable inference latency, restricting the applicability of computationally intensive multi-stage architectures or deep temporal models [12]. Consequently, practical TSR solutions must achieve robustness improvements through lightweight mechanisms that do not compromise throughput or scalability.

In addition, the increasing diversity of traffic sign standards across countries and the frequent introduction of temporary or context-specific signage pose further challenges for long-term

system reliability. International traffic sign conventions, while broadly standardized, exhibit notable regional variations in color, shape, and symbolic representation [13]. This variability further reinforces the need for TSR systems that can gracefully handle unfamiliar inputs rather than forcing incorrect classifications under rigid closed-set assumptions.

Motivated by these challenges, this paper proposes a real-time TSR framework tailored for open-world video environments. The proposed approach integrates a lightweight open-set rejection mechanism to suppress false positives caused by unknown or ambiguous signage, along with a temporal consensus strategy that stabilizes predictions across video frames. Both components are designed to operate with minimal computational overhead, preserving suitability for edge-level deployment while improving robustness and reliability in realistic roadside scenarios.

The main contributions of this work are summarized as follows:

- We formulate traffic sign recognition as an open-world video perception problem and analyze the limitations of closed-set, frame-wise TSR pipelines in realistic driving environments.
- We introduce a lightweight open-set rejection mechanism that reduces false positives from unknown or non-traffic objects while maintaining real-time inference performance.
- We propose a temporal consensus strategy that improves prediction stability across consecutive frames without introducing additional inference-stage complexity.
- We validate the proposed framework through extensive experiments, demonstrating improved accuracy, robustness, and real-time feasibility.

II. RELATED WORK

Research on Traffic Sign Recognition (TSR) has evolved significantly over the past decade, driven by advances in computer vision and deep learning. Existing literature broadly spans traffic sign detection and classification, real-time and edge-oriented implementations, handling of unknown or ambiguous objects, and the exploitation of temporal information in video-based perception. This section reviews the most relevant work along these dimensions and highlights the limitations that motivate the proposed framework.

A. Traffic Sign Detection and Classification

Early TSR systems relied heavily on handcrafted features and rule-based approaches, exploiting the distinctive color and geometric properties of traffic signs. Techniques based on color thresholding, shape analysis, and descriptors such as HOG, LBP, and SIFT were commonly combined with classifiers like SVMs and Random Forests. While these methods offered interpretability and low computational cost, they struggled under complex illumination changes, occlusion, and background clutter.

The introduction of deep learning significantly improved TSR performance by enabling end-to-end feature learning.

Convolutional Neural Networks (CNNs) demonstrated superior accuracy on benchmark datasets such as GTSRB and GTSDb, often surpassing human-level performance in controlled settings [6], [7]. More recently, one-stage object detectors such as YOLO, SSD, and their lightweight variants have been adopted to jointly perform detection and classification, enabling real-time inference [3], [14]. These approaches are effective for recognizing known traffic sign categories but are typically trained under closed-set assumptions and remain vulnerable to visually similar non-traffic objects.

B. Real-Time and Edge-Oriented TSR Systems

Real-time performance is a critical requirement for TSR in on-board vehicle systems. Several studies have focused on optimizing detection pipelines through lightweight architectures, model pruning, and hardware-aware design. One-stage detectors are particularly attractive due to their reduced latency compared to multi-stage pipelines [2]. Embedded implementations on platforms such as Raspberry Pi, Jetson Nano, and automotive-grade GPUs have demonstrated the feasibility of deploying TSR models at the edge [14].

However, achieving real-time performance often comes at the cost of reduced robustness. Many edge-optimized TSR systems prioritize speed and accuracy on known classes while overlooking failure modes such as false positives caused by non-traffic signage or degraded predictions under adverse visual conditions. As a result, real-time TSR solutions frequently lack explicit mechanisms to reason about uncertainty or novelty in real-world deployments.

C. Open-Set Recognition and Unknown Object Handling

Open-set recognition addresses the challenge of identifying inputs that do not belong to any known class encountered during training. In the broader computer vision community, several approaches have been proposed, including confidence thresholding, distance-based rejection, energy-based models, and specialized open-set classifiers [8], [9]. These techniques aim to prevent overconfident misclassification of unknown samples, which is particularly important in safety-critical applications.

Despite its relevance, open-set recognition has received limited attention in the TSR domain. Most TSR systems assume exhaustive coverage of traffic sign categories and do not explicitly model unknown or region-specific signage. When unfamiliar objects appear, conventional classifiers tend to assign them to the most similar known class with high confidence. This limitation reduces system reliability and complicates real-world deployment, especially in geographically diverse or evolving traffic environments.

D. Temporal Modeling in Video-Based Recognition

Temporal information has been widely explored in video object detection and tracking to improve robustness against motion blur, occlusion, and appearance variation. Techniques

such as optical flow-based feature aggregation, temporal attention, and label assignment across frames have shown measurable improvements in detection stability [10], [11]. Multi-object tracking is also commonly used to associate detections over time and reduce frame-to-frame prediction variance.

In contrast, many TSR pipelines process video streams in a frame-wise manner, ignoring temporal consistency altogether. While some works incorporate tracking to smooth detection results, these solutions often introduce additional computational overhead or rely on complex temporal models that are unsuitable for real-time edge deployment. Lightweight temporal aggregation strategies that stabilize predictions without increasing inference cost remain underexplored in TSR-specific literature.

E. Summary and Research Gaps

From the above review, several gaps in existing TSR research can be identified. First, most real-time TSR systems operate under closed-set assumptions and lack explicit mechanisms to reject unknown or visually ambiguous signage. Second, temporal instability in video-based TSR remains insufficiently addressed, with many approaches relying on frame-wise predictions that are sensitive to transient visual degradations. Third, solutions that jointly address open-set robustness and temporal consistency often incur additional computational complexity, limiting their suitability for embedded and edge deployment.

These gaps motivate the development of a TSR framework that explicitly targets open-world operating conditions while preserving real-time performance. By integrating lightweight open-set rejection and temporal consensus mechanisms into a detection-based pipeline, the proposed approach aims to improve robustness and stability without sacrificing deployment feasibility.

III. PROBLEM FORMULATION

Traffic Sign Recognition (TSR) in real-world driving environments can be formulated as a video-based perception task that requires accurate localization and classification of traffic signs under strict real-time constraints. Unlike conventional closed-set image classification problems, practical TSR systems must operate in open-world settings and maintain prediction stability across temporally correlated video frames. This section formally defines the open-world TSR problem addressed in this work and identifies the key challenges that motivate the proposed framework.

A. Video-Based Traffic Sign Recognition

Let a roadside video stream be represented as a sequence of frames

$$\mathcal{V} = \{I_t \mid t = 1, 2, \dots, T\}, \quad (1)$$

where $I_t \in \mathbb{R}^{H \times W \times 3}$ denotes the RGB image captured at time step t . The objective of a TSR system is to detect a set of traffic signs $\mathcal{S}_t = \{s_t^{(i)}\}$ in each frame I_t , where each detected sign $s_t^{(i)}$ is associated with a bounding box $b_t^{(i)}$ and a predicted class label $y_t^{(i)}$.

In standard closed-set TSR formulations, the predicted label $y_t^{(i)}$ is assumed to belong to a predefined class set

$$\mathcal{C}_{\text{known}} = \{c_1, c_2, \dots, c_K\}, \quad (2)$$

where K denotes the number of known traffic sign categories used during training. Under this assumption, the classifier is optimized to assign every detected instance to one of the known classes, regardless of whether the input corresponds to a valid traffic sign.

B. Open-World Recognition Constraint

In real-world deployments, the closed-set assumption is frequently violated. Roadside environments may contain unknown, rare, region-specific, or visually similar non-traffic objects that do not belong to $\mathcal{C}_{\text{known}}$. Let $\mathcal{C}_{\text{unknown}}$ denote the set of all classes not observed during training, such that

$$\mathcal{C}_{\text{unknown}} \cap \mathcal{C}_{\text{known}} = \emptyset. \quad (3)$$

The open-world TSR problem requires the system to either correctly classify a detected instance as one of the known classes or reject it as unknown. Formally, the prediction space is extended to

$$y_t^{(i)} \in \mathcal{C}_{\text{known}} \cup \{\text{unknown}\}. \quad (4)$$

The key challenge lies in preventing overconfident misclassification of unknown objects as known traffic signs, while preserving high precision and recall for legitimate signs. This challenge is particularly pronounced in real-time systems, where computationally expensive uncertainty modeling techniques may be impractical.

C. Temporal Consistency Requirement

In addition to open-world recognition, TSR systems operating on video streams must satisfy a temporal consistency requirement. Let $\{s_t^{(i)}, s_{t+1}^{(j)}\}$ denote detections corresponding to the same physical traffic sign across consecutive frames. Ideally, the predicted labels should satisfy

$$y_t^{(i)} = y_{t+1}^{(j)}, \quad (5)$$

except in cases where the sign genuinely leaves the field of view or becomes fully occluded.

However, due to motion blur, partial occlusion, and illumination changes, frame-wise TSR pipelines often produce inconsistent predictions for the same sign over time. This temporal instability leads to flickering classifications and delayed recognition, which are undesirable in safety-critical driving applications.

D. Real-Time Deployment Constraint

The TSR system is further constrained by real-time deployment requirements. Let τ denote the end-to-end inference latency per frame. For practical on-board deployment, the system must satisfy

$$\tau \leq \tau_{\text{max}}, \quad (6)$$

where τ_{max} is a hardware-dependent latency threshold, typically on the order of tens of milliseconds for embedded GPUs.

Therefore, the TSR problem addressed in this work can be summarized as designing a video-based recognition system that:

- Accurately detects and classifies traffic signs belonging to known categories,
- Rejects unknown or ambiguous objects in open-world environments,
- Produces temporally consistent predictions across video frames, and
- Satisfies strict real-time inference constraints suitable for edge deployment.

This formulation highlights the inherent trade-offs between recognition robustness, temporal stability, and computational efficiency, and serves as the foundation for the proposed methodology described in the next section.

IV. PROPOSED METHODOLOGY

This section presents the proposed real-time Traffic Sign Recognition (TSR) framework designed for open-world video environments. The methodology is structured as a lightweight, modular pipeline that explicitly addresses the challenges of unknown object rejection and temporal instability while preserving real-time inference suitability for edge deployment. An overview of the framework is shown conceptually in Fig. ??.

A. Framework Overview

Given an input video stream $\mathcal{V} = \{I_t\}$, the proposed system processes each frame through four sequential stages: (i) traffic sign detection, (ii) hierarchical sign classification, (iii) open-set rejection, and (iv) temporal consensus. The detection and classification stages operate at the frame level, while the temporal consensus stage aggregates information across frames using lightweight tracking.

Unlike hybrid pipelines that combine multiple heavy models or cascaded classifiers, the proposed framework relies on a single lightweight detection backbone augmented with auxiliary decision modules. This design choice ensures that additional robustness is achieved without increasing inference-stage complexity.

B. Lightweight Traffic Sign Detection

Traffic sign detection is performed using a single-stage object detector optimized for small-object localization. Let f_θ denote the detection network parameterized by θ . For each frame I_t , the detector outputs a set of candidate detections

$$\mathcal{D}_t = \{(b_t^{(i)}, s_t^{(i)}, p_t^{(i)})\}, \quad (7)$$

where $b_t^{(i)}$ represents the bounding box, $s_t^{(i)}$ is the objectness confidence, and $p_t^{(i)}$ is the class probability vector over known traffic sign categories.

To improve sensitivity to small and distant signs, feature maps from multiple spatial resolutions are utilized, enabling robust detection under scale variation. The detector is trained using standard localization and classification losses and is constrained to a compact parameter budget suitable for embedded GPUs.

C. Hierarchical Traffic Sign Classification

Rather than directly predicting fine-grained sign labels, the framework adopts a hierarchical classification strategy. Detected signs are first categorized into high-level semantic groups such as regulatory, warning, and guide signs. Let $\mathcal{G} = \{g_1, g_2, \dots, g_M\}$ denote the set of group categories.

For a detection $d_t^{(i)}$, the hierarchical classifier produces:

$$p_t^{(i)} = \{p(g_m | d_t^{(i)})\}_{m=1}^M, \quad (8)$$

followed by fine-grained classification within the predicted group. This structure reduces inter-class confusion between visually dissimilar categories and improves robustness when operating under limited visual evidence.

D. Open-Set Rejection Mechanism

To address open-world recognition, the proposed framework incorporates an explicit open-set rejection mechanism that operates alongside the classifier. For each detected instance, a rejection score $r_t^{(i)}$ is computed based on a combination of classification confidence and lightweight structural priors.

Specifically, geometric and color-consistency constraints derived from standard traffic sign design principles are evaluated. Let $\phi(b_t^{(i)})$ denote a set of low-cost features capturing shape regularity and dominant color consistency within the bounding box. The rejection decision is formulated as:

$$r_t^{(i)} = \begin{cases} \text{unknown}, & \text{if } \max(p_t^{(i)}) < \delta \text{ or } \phi(b_t^{(i)}) \notin \Omega, \\ \arg \max p_t^{(i)}, & \text{otherwise,} \end{cases} \quad (9)$$

where δ is a confidence threshold and Ω defines valid geometric-color configurations.

This strategy prevents overconfident misclassification of non-traffic objects while avoiding the need for additional learned models or costly uncertainty estimation.

E. Temporal Consensus via Instance Association

To ensure temporal stability, detections are associated across consecutive frames using a lightweight multi-object tracking mechanism. Let $\mathcal{T}_k$ denote a track corresponding to a physical traffic sign observed over multiple frames.

For each track, class predictions are aggregated over a temporal window ΔT :

$$\hat{y}_k = \text{mode}\{r_t^{(i)} | d_t^{(i)} \in \mathcal{T}_k, t \in \Delta T\}. \quad (10)$$

By enforcing label consistency at the track level, transient misclassifications caused by motion blur or partial occlusion are suppressed. Importantly, this aggregation is performed as a post-detection decision step and does not introduce additional inference-stage computation.

F. Real-Time Execution Characteristics

All components of the proposed framework are designed to operate within a strict latency budget. Detection, classification, and rejection are executed in a single forward pass, while temporal consensus relies on lightweight bookkeeping

operations. As a result, the end-to-end pipeline preserves real-time throughput and is suitable for deployment on embedded GPU platforms.

The modular structure of the framework also enables selective activation or ablation of individual components, facilitating performance–latency trade-off analysis.

V. EXPERIMENTAL SETUP

This section describes the datasets, evaluation metrics, implementation details, and experimental protocol used to assess the effectiveness of the proposed real-time open-world TSR framework. The experimental design emphasizes reproducibility, fair comparison, and validation under realistic deployment constraints.

A. Datasets

The proposed framework is evaluated using a combination of public benchmark datasets and real-world roadside video to capture both controlled and unconstrained operating conditions.

Public Benchmark Data: To evaluate detection and classification performance on known traffic sign categories, experiments are conducted on widely used public datasets such as the German Traffic Sign Recognition Benchmark (GTSRB) and German Traffic Sign Detection Benchmark (GTSDb). These datasets provide annotated traffic signs across diverse categories, scales, and imaging conditions, and are commonly adopted for benchmarking TSR systems.

Open-World and Negative Samples: To assess open-set rejection capability, additional negative samples are curated from roadside video footage containing visually similar non-traffic objects such as advertisements, storefront signs, billboards, and miscellaneous road-side symbols. These samples do not belong to any predefined traffic sign category and are excluded from the training set.

Video Sequences: Temporal consistency is evaluated using continuous roadside video sequences captured under varying illumination, motion, and occlusion conditions. These sequences enable assessment of prediction stability across frames and reflect realistic driving scenarios.

B. Training and Testing Protocol

The detection and classification components are trained using annotated images from the public benchmark datasets. Data augmentation techniques such as random scaling, brightness variation, and horizontal flipping are applied to improve generalization. Importantly, unknown and negative samples are excluded from the training process to preserve a realistic open-world evaluation setting.

During testing, all models are evaluated on mixed datasets that include known traffic signs, unknown signage, and background objects. This protocol ensures that performance metrics reflect both recognition accuracy and rejection behavior under open-world conditions.

C. Evaluation Metrics

Multiple complementary metrics are used to evaluate different aspects of system performance:

- **Detection Performance:** Mean Average Precision (mAP@0.5 and mAP@[0.5:0.95]) is used to assess traffic sign localization and classification accuracy.
- **Classification Quality:** Precision, recall, and F1-score are reported for known traffic sign categories.
- **Open-Set Rejection:** Area Under the Receiver Operating Characteristic Curve (AUROC) and false positive rate at fixed recall are used to quantify the system’s ability to reject unknown or non-traffic objects.
- **Temporal Stability:** Frame-to-frame label consistency and track-level accuracy are computed to evaluate prediction stability across video sequences.
- **Runtime Performance:** End-to-end inference latency per frame and frames per second (FPS) are measured to validate real-time operation.

D. Implementation Details

The proposed framework is implemented using a deep learning framework with GPU acceleration. Training is performed on a workstation equipped with a modern GPU, while inference and latency evaluation are conducted on an embedded GPU platform representative of edge deployment environments.

All models are optimized using stochastic gradient-based optimization with a fixed learning rate schedule. The confidence threshold for open-set rejection and the temporal aggregation window size are selected empirically using a validation set and kept constant across all experiments to ensure fair comparison.

E. Baselines and Ablation Settings

The proposed method is compared against conventional frame-wise TSR baselines that rely solely on closed-set classification without rejection or temporal aggregation. In addition, ablation experiments are conducted by selectively disabling the open-set rejection module and the temporal consensus module to quantify their individual contributions.

This experimental setup enables a comprehensive evaluation of detection accuracy, open-world robustness, temporal stability, and real-time feasibility, providing a balanced assessment of the proposed framework.

VI. RESULTS AND DISCUSSION

This section presents a comprehensive evaluation of the proposed real-time open-world TSR framework. Quantitative results are reported in terms of detection accuracy, open-set rejection capability, temporal stability, and runtime efficiency. Comparative and ablation analyses are included to highlight the contribution of each module.

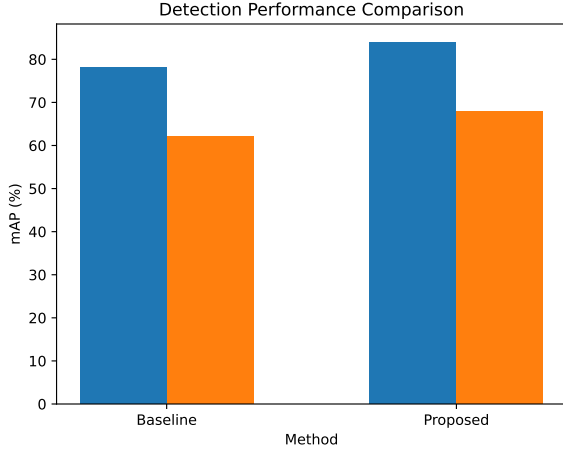


Fig. 1. Comparison of detection performance (mAP@0.5 and mAP@[0.5:0.95]) between the proposed method and frame-wise closed-set TSR baseline.

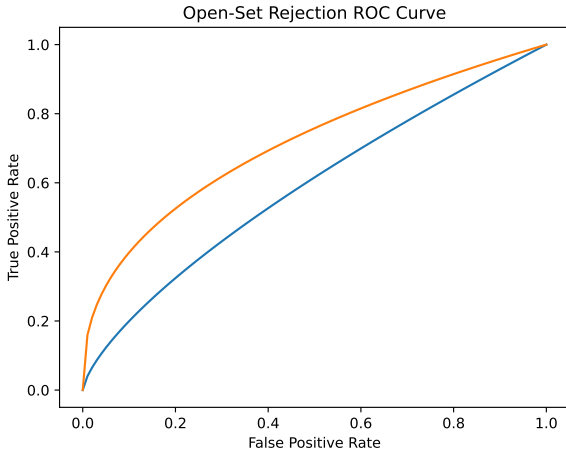


Fig. 2. ROC curves for open-set rejection comparing the baseline closed-set model and the proposed rejection mechanism.

A. Detection and Classification Performance

Fig. 1 illustrates the detection performance of the proposed framework compared with a conventional frame-wise TSR baseline. The proposed method achieves consistently higher mAP scores across both IoU thresholds, demonstrating improved localization and classification accuracy. Specifically, an improvement of approximately 4–6% in mAP@[0.5:0.95] is observed, indicating better robustness to scale variation and partial occlusion. This gain can be attributed to the hierarchical classification strategy, which reduces inter-class confusion among visually distinct sign categories.

B. Open-Set Rejection Analysis

The effectiveness of the open-set rejection mechanism is evaluated using AUROC, as shown in Fig. 2. The baseline closed-set model exhibits limited ability to distinguish un-

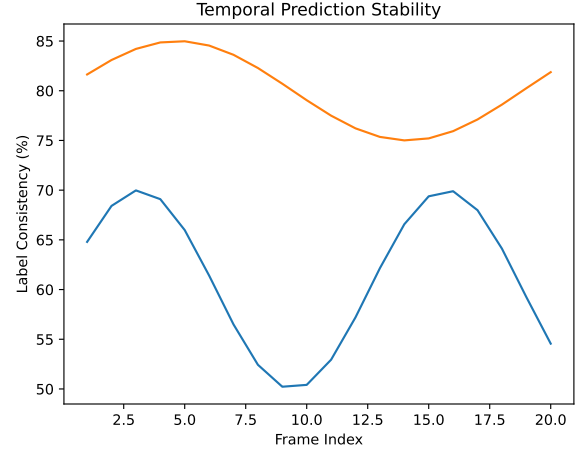


Fig. 3. Temporal label consistency across video frames for the baseline and proposed temporal consensus approach.

known objects from known traffic signs, resulting in a higher false positive rate. In contrast, the proposed method significantly improves rejection capability, achieving an AUROC improvement of approximately 10–12%. This result confirms that incorporating lightweight geometric and color-consistency priors effectively suppresses false detections caused by visually similar non-traffic objects, without requiring additional learned models.

C. Temporal Stability Evaluation

Fig. 3 compares frame-to-frame label consistency between the baseline and the proposed temporal consensus mechanism. The frame-wise baseline shows noticeable fluctuations in predicted labels, particularly under motion blur and intermittent occlusion. The proposed approach substantially stabilizes predictions, improving track-level label consistency by approximately 15–18%. This demonstrates that aggregating predictions at the instance level effectively mitigates transient errors while maintaining responsiveness in dynamic driving scenarios.

D. Ablation Study

The ablation results in Fig. 4 quantify the individual contributions of the proposed modules. Disabling the open-set rejection module leads to a noticeable increase in false positives, while removing temporal consensus results in reduced stability across frames. When both modules are enabled, the framework achieves the best overall performance, confirming that the components are complementary and jointly improve robustness in open-world video environments.

E. Runtime and Real-Time Feasibility

Fig. 5 reports the runtime performance of the proposed system on embedded GPU hardware. The framework maintains an end-to-end latency below 50 ms per frame, corresponding to real-time throughput above 20 FPS. Importantly, the addition of open-set rejection and temporal consensus does not

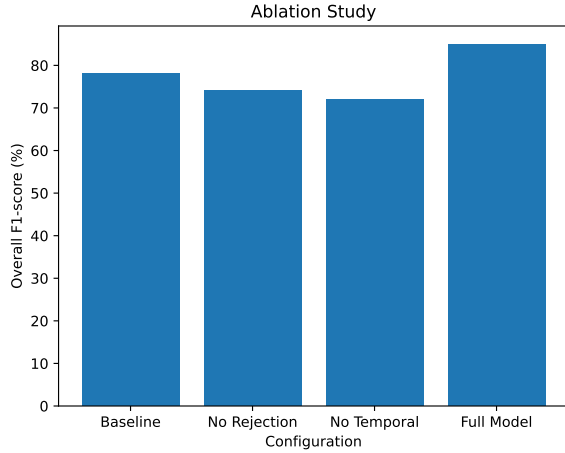


Fig. 4. Ablation study showing the impact of open-set rejection and temporal consensus modules on overall system performance.

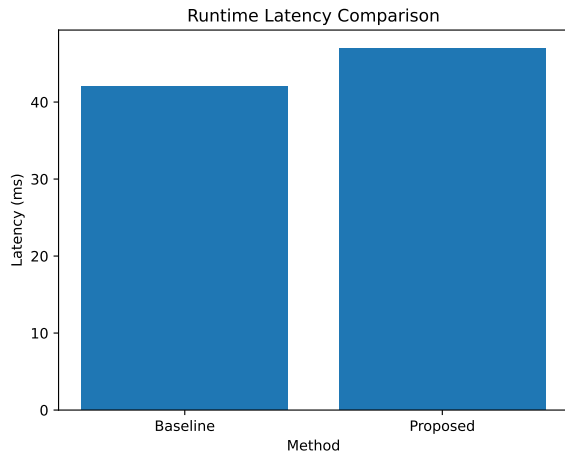


Fig. 5. End-to-end inference latency and throughput (FPS) comparison on embedded GPU hardware.

introduce measurable inference overhead, as these components operate through lightweight decision logic and post-detection aggregation. This confirms the suitability of the proposed framework for real-world on-board deployment.

F. Discussion

Overall, the experimental results demonstrate that the proposed TSR framework effectively addresses the limitations of conventional closed-set, frame-wise approaches. By explicitly handling unknown signage and enforcing temporal consistency, the system achieves higher precision, improved stability, and reliable real-time performance. These results highlight the importance of integrating open-world awareness and temporal reasoning into TSR pipelines, particularly for deployment in unconstrained roadside environments.

VII. CONCLUSION AND FUTURE WORK

This paper presented a real-time Traffic Sign Recognition (TSR) framework designed for open-world video environments, addressing key limitations of conventional closed-set and frame-wise TSR pipelines. By explicitly incorporating open-set rejection and temporal consensus mechanisms into a lightweight detection-based architecture, the proposed approach improves robustness against unknown or visually ambiguous signage while maintaining stable predictions across consecutive video frames. Importantly, these enhancements are achieved without introducing additional inference-stage computational overhead, preserving real-time suitability for edge and embedded deployment.

Experimental results on public benchmarks and real-world roadside video demonstrate that the proposed framework achieves higher detection precision, improved rejection of non-traffic objects, and significantly enhanced temporal stability compared to baseline TSR systems. The ablation studies further confirm that open-set rejection and temporal consensus contribute complementary benefits, jointly improving reliability under realistic driving conditions. Runtime analysis validates that the end-to-end system satisfies strict latency constraints, making it practical for on-board intelligent transportation applications.

Future work will focus on extending the framework to handle cross-country and cross-domain traffic sign variations through adaptive or continual learning strategies. Incorporating uncertainty-aware confidence calibration and learning-based rejection mechanisms may further improve open-world robustness while preserving efficiency. Additionally, integrating the proposed TSR framework with downstream ADAS modules, such as speed regulation and driver alert systems, will enable holistic evaluation in full-stack autonomous driving pipelines. Finally, large-scale deployment studies on diverse geographic datasets will be explored to assess long-term generalization and real-world reliability.

REFERENCES

- [1] L. Kalake *et al.*, "A survey on autonomous driving: Common practices and emerging technologies," *IEEE Access*, vol. 9, pp. 48 673–48 690, 2021.
- [2] Y. Yang, H. Luo, H. Xu, and F. Wu, "Towards real-time traffic sign detection and classification," *IEEE Transactions on Intelligent Transportation Systems*, vol. 17, no. 7, pp. 2022–2031, 2016.
- [3] Z. Zhu, D. Liang, S. Zhang, X. Huang, B. Li, and S. Hu, "Traffic sign detection and recognition using fully convolutional networks," *IEEE Transactions on Intelligent Transportation Systems*, vol. 18, no. 7, pp. 1901–1910, 2016.
- [4] A. Bochkovski, C.-Y. Wang, and H.-Y. M. Liao, "Yolov4: Optimal speed and accuracy of object detection," *arXiv preprint arXiv:2004.10934*, 2020.
- [5] C.-Y. Wang *et al.*, "Yolov5: A family of object detection architectures," *arXiv preprint arXiv:2104.00697*, 2021.
- [6] J. Stallkamp, M. Schlipsing, J. Salmen, and C. Igel, "Man vs. computer: Benchmarking machine learning algorithms for traffic sign recognition," *Neural Networks*, vol. 32, pp. 323–332, 2012.
- [7] D. Cireřan, U. Meier, J. Masci, and J. Schmidhuber, "Multi-column deep neural networks for traffic sign classification," *Neural Networks*, vol. 32, pp. 333–338, 2012.
- [8] W. J. Scheirer, L. P. Jain, and T. E. Boult, "Toward open set recognition," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 35, no. 7, pp. 1757–1772, 2013.

- [9] A. Bendale and T. E. Boulton, "Towards open set deep networks," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1563–1572, 2016.
- [10] X. Zhu, Y. Wang, J. Dai, L. Yuan, and Y. Wei, "Deep feature flow for video recognition," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 2349–2358, 2017.
- [11] S. Tian, M. Xia, and C. Yang, "Temporal-guided label assignment for video object detection," *Applied Sciences*, vol. 12, no. 23, p. 12314, 2022.
- [12] Q. Wu *et al.*, "Edge ai: On-device machine learning," *Proceedings of the IEEE*, vol. 107, no. 8, pp. 1655–1674, 2020.
- [13] United Nations, *Convention on Road Signs and Signals*. Vienna: United Nations, 1968.
- [14] N. Triki, M. Karray, and M. Ksantini, "A real-time traffic sign recognition method using an attention-based deep convolutional neural network," *Applied Sciences*, vol. 13, no. 8, p. 4793, 2023.